FairLight: Fairness-Aware City-Wide Emergency Vehicle Prioritization via Hierarchical Multi-Agent Reinforcement Learning

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Abstract

Urban traffic signal control is a complex sequential decision-making problem in which decisions made at one intersection can influence traffic conditions at neighboring and downstream intersections. The problem becomes increasingly challenging as the size of the road network grows because a controller must simultaneously achieve network-level coordination, local responsiveness, and computational scalability. Reinforcement learning has emerged as a promising approach for adaptive traffic signal control, while multi-agent reinforcement learning (MARL) allows control to be distributed among individual intersections. However, fully decentralized approaches may lack sufficient global context, whereas centralized approaches become difficult to scale.

Hierarchical MARL frameworks such as HiLight reduce this problem to a Meta-Policy/Sub-Policy hierarchy coordinated over a fixed, manually preprocessed region partition and a constant per-intersection neighborhood, and do not model emergency-vehicle (EMV) priority or the fairness cost that priority imposes on ordinary traffic. This paper presents FairLight, a Hierarchical Region-Aware Multi-Agent Reinforcement Learning framework that (i) replaces fixed regional preprocessing with automatic, graph-based partitioning — Louvain and Leiden community detection over a traffic-weighted intersection graph, spatial DBSCAN, and four hybrid combinations of the two — evaluated as an experimental variable rather than assumed fixed; (ii) augments each intersection's local observation with a graph-attention neighborhood representation capped at four neighbors (truncated or zero-padded as needed, not an arbitrary-degree mechanism); and (iii) adds an explicit fairness penalty to the Sub-Policy reward, combining per-intersection waiting-time variance, a starvation threshold, and Jain's Fairness Index, so that aggregate efficiency gains cannot be reported without also reporting their distributional cost to ordinary vehicles.

The framework is implemented and trained with Proximal Policy Optimization and Generalized Advantage Estimation on top of a Transformer/LSTM Meta-Policy and a graph-attention Sub-Policy, and is configured for five SUMO traffic networks (a 472-intersection Manhattan extract, the Cologne8, Ingolstadt21, Arterial4x4, and Grid4x4 benchmark networks). City-wide coordination of multiple simultaneously active EMVs with conflict arbitration, and incident/accident-robust training, are presented as proposed extensions to this architecture (Sections 3.2 and 3.3): no emergency vehicles are present in any of the five scenarios' current demand data, and no corresponding training or evaluation code has been implemented; a formal envy-free objective is likewise not implemented, and the fairness mechanism above should not be read as an envy-free guarantee. What has been measured is the automatic regionalization step itself: applied to the five scenarios, Louvain and Leiden produce between 1 and 70 regions depending on network topology, with modularity ranging from near zero on small, sparsely connected networks to above 0.6 on Manhattan, while DBSCAN's purely spatial clustering yields coarser, near-uniform partitions across all five. Full closed-loop evaluation of the trained controller against Fixed-Time and Max-Pressure baselines has not yet been completed at the time of writing; the corresponding results are reported as pending rather than estimated. This manuscript is accordingly a methodological and partially evaluated contribution: it documents what is implemented and measured, and separates these findings explicitly from what remains proposed future work.

Keywords: Traffic Signal Control, Multi-Agent Reinforcement Learning, Emergency Vehicle Routing, Community Detection, Louvain Algorithm, Fairness-Aware Optimization, Hierarchical Reinforcement Learning

1. Introduction

Urban traffic signal control is a complex sequential decision-making problem where decisions at one intersection propagate to neighboring and downstream intersections. Traffic demand varies dynamically with time, incidents, weather, and travel patterns, making static or rule-based controllers insufficient for real-world conditions. Adaptive systems such as SCOOT introduced traffic-responsive coordination [2], while reinforcement learning approaches have since enabled data-driven signal control [11]. Multi-agent reinforcement learning (MARL) further distributes control across intersections, improving scalability, though fully decentralized approaches risk optimizing locally at the expense of network-wide coordination [20, 19].

Beyond aggregate congestion, traffic signal decisions simultaneously affect travel time, delay, queue formation, throughput, fuel consumption, and emissions. Critically, evaluating controllers solely through aggregate metrics such as average waiting time conceals how service is distributed among individual vehicles. Consider two controllers with vehicle waiting times $W_A = [10, 10, 10, 10, 10, 100]$ and $W_B = [25, 25, 25, 25, 25, 25]$: Controller A achieves a lower mean delay yet inflicts severe starvation on one vehicle. This motivates fairness as a first-class control objective rather than a post-hoc evaluation criterion.

A further dimension of real-world uncertainty that existing MARL frameworks largely overlook is the presence of emergency vehicles (EMVs). Most traffic signal control approaches, including hierarchical methods such as HiLight [24] and cooperative approaches such as CoLight [17], assume homogeneous vehicle populations and optimize purely for aggregate efficiency. In practice, ambulances, fire engines, and police vehicles impose stochastic, high-priority demands on the network that cannot be resolved by ordinary signal timing. Delayed EMV passage has measurable consequences: studies report approximately 700 annual fatalities attributable to late ambulance responses, and an estimated annual mean of 4,500 ambulance-involved crashes in the United States [35]. EMVLight demonstrated that jointly optimizing EMV routing and adaptive signal control within a multi-agent actor-critic framework substantially reduces emergency response times [31], but does not address hierarchical city-wide coordination or fairness. These considerations establish EMV responsiveness as a necessary component of any realistic traffic management framework, not an optional extension. At the same time, providing unrestricted priority to an EMV may excessively delay ordinary traffic, potentially transferring congestion to other parts of the network rather than resolving it. EMV prioritization should therefore not be treated as an isolated intersection-level problem: it requires coordination across the surrounding traffic network while maintaining acceptable service for non-emergency vehicles.

Hierarchical reinforcement learning addresses the scalability–coordination tradeoff by separating long-horizon strategic decisions from short-horizon local control [22, 23]. HiLight is particularly relevant here, employing a Meta-Policy with Transformer-based spatial and LSTM-based temporal modelling to coordinate decentralized Sub-Policy agents across large networks [24]. As reported by its authors, its region partitioning is a preprocessing step rather than a control-relevant variable, and it does not incorporate EMV priority or vehicle-level fairness; we treat this characterization of HiLight as the motivating baseline throughout this paper rather than as a re-verified, independently reproduced claim.

This paper presents FairLight, a Hierarchical Multi-Agent Actor–Critic framework that investigates three extensions to this class of baseline: (1) replacing a fixed or manually specified region partition with automatic, graph-based regionalization, evaluated as an experimental variable rather than a fixed preprocessing choice; (2) an explicit, measured fairness penalty on ordinary-vehicle waiting-time disparity, in place of a purely aggregate efficiency objective; and (3) a proposed design, not yet implemented, for coordinating multiple simultaneously active emergency vehicles with conflicting routes, motivated by the observation that single-EMV preemption strategies degrade ordinary-traffic efficiency and do not extend trivially to concurrent EMVs. The paper's central research questions are: (RQ1) does the choice of automatic regionalization method materially change the region structure handed to the Meta-Policy, and by how much, across networks of different size and topology; (RQ2) can an explicit fairness penalty be added to the Sub-Policy reward without requiring a separate architecture; and (RQ3) what would be required, beyond the existing architecture, to support city-wide multi-EMV coordination. RQ1 is answered empirically in Section 6 using five SUMO networks; RQ2 is answered at the level of reward design and implementation (Section 4), with closed-loop evaluation pending; RQ3 is addressed as a design proposal (Section 3.2) rather than a validated result, since no part

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of the multi-EMV mechanism is currently implemented and none of the five evaluation scenarios currently include an emergency vehicle in their demand data.

The contributions of this paper, each labeled by its evidentiary status, are:

1. **Implemented and measured.** Automatic regionalization of the signalized road network via Louvain and Leiden community detection over a traffic-weighted intersection graph, spatial DBSCAN, and four hybrid compositions of the two, with region counts, cluster sizes, and modularity reported for five networks (Section 6).
2. **Implemented, evaluation pending.** A hierarchical Meta-Policy/Sub-Policy architecture trained end-to-end with Proximal Policy Optimization, in which the Sub-Policy reward includes an explicit fairness penalty (waiting-time variance, a starvation threshold, and Jain's Fairness Index) in addition to local traffic efficiency (Sections 3–4).
3. **Implemented, scope-limited.** A graph-attention neighborhood mechanism that incorporates information from up to four neighboring intersections per decision step, with truncation or zero-padding outside that range (Section 3.7).
4. **Proposed, not implemented.** A city-wide multi-EMV routing and signal-priority arbitration mechanism for simultaneously active emergency vehicles (Section 3.2), and an incident/accident-robustness training mechanism (Section 3.3).

To the best of our knowledge, no existing framework combines automatic, experimentally-compared regional partitioning with an explicit ordinary-traffic fairness penalty inside a hierarchical MARL controller; the multi-EMV and formal fairness-guarantee extensions remain future work rather than demonstrated contributions.

2. Related Work

Conventional traffic signal control has primarily relied on analytical optimization and rule-based methods for determining optimal signal timings. Webster's signal timing formulation introduced one of the earliest mathematical models for fixed-cycle traffic signal optimization, while later adaptive traffic control systems such as SCOOT and SCATS dynamically adjusted signal timings using detector-based traffic measurements and predefined control strategies [1, 2, 3, 4]. Max-pressure control further improved network throughput by selecting signal phases according to queue pressure differences between incoming and outgoing links [5]. Although these approaches are computationally efficient and have been successfully deployed in practice, they rely heavily on predefined traffic models and handcrafted optimization rules. Consequently, they struggle to adapt to highly dynamic urban environments involving unpredictable congestion, heterogeneous intersections, accidents, and emergency vehicle requests.

To improve adaptability, several optimization and machine learning techniques have been introduced for traffic signal control. Evolutionary optimization methods such as Genetic Algorithms optimize signal timing plans through iterative search, while Fuzzy Logic Controllers employ expert-defined rules to respond to uncertain traffic conditions [6, 7]. Later studies adopted machine learning techniques including Support Vector Machines, Artificial Neural Networks, and Long Short-Term Memory (LSTM) networks to predict traffic flow and congestion using historical observations [8, 9, 10]. Although these methods improve prediction accuracy and optimization quality, they primarily estimate future traffic states rather than learning sequential traffic control policies. Furthermore, they require extensive historical training data and cannot continuously adapt to rapidly changing traffic conditions.

Recent advances in deep reinforcement learning have significantly improved adaptive traffic signal control by enabling traffic controllers to learn directly through interaction with the traffic environment. IntelliLight demonstrated the effectiveness of Deep Q-Networks for adaptive traffic signal control at isolated intersections [11]. PressLight integrated max-pressure traffic theory into deep reinforcement learning to maximize traffic throughput while reducing handcrafted reward engineering [12]. FRAP modeled traffic control as a phase competition problem and exploited traffic phase symmetry to improve learning efficiency [13]. AttendLight later introduced an attention-based architecture capable of generalizing across heterogeneous intersections with varying lane configurations [14]. MetaLight focused on transferring learned policies across unseen intersections [15], while Advanced-XLight demonstrated that effective traffic state representation can significantly influence performance in reinforcement-learning-based traffic signal control [16]. Despite these advances, existing deep reinforcement learning methods primarily optimize local traffic efficiency and do not jointly consider emergency vehicle prioritization, fairness, or large-scale hierarchical coordination.

To improve cooperation among multiple intersections, graph neural networks and multi-agent reinforcement learning have received increasing attention. CoLight introduced graph attention networks to enable communication among neighboring intersections for cooperative traffic signal control [17]. MPLight further improved scalability by introducing parameter sharing across reinforcement learning agents, allowing efficient deployment over city-scale transportation networks [18]. HiLight extended this concept by proposing a hierarchical reinforcement learning framework consisting of Meta-Policy and Sub-Policy agents for coordinating thousands of intersections efficiently [24]. CityFlow provides a large-scale multi-agent reinforcement learning environment designed specifically for city traffic scenarios and supports evaluation of scalable traffic-control algorithms [21]. Although these methods substantially improve scalability and inter-agent cooperation, their primary objective remains maximizing traffic efficiency. They neither address emergency vehicle routing nor incorporate fairness-aware resource allocation across different regions of a city.

HiLight's hierarchical Meta-Policy/Sub-Policy formulation [24], like most of the methods discussed above, operates under a closed-world assumption: it is trained and evaluated on recurrent congestion patterns and does not model non-recurrent disruptions such as emergency vehicle preemption or traffic incidents. This omission is not merely conceptual. Poorly coordinated EMV preemption has been shown to increase average delay for cross-traffic by 18–32 seconds per vehicle per preemption event, with these localized delays cascading into network-wide congestion [32]. Even well-established preemption strategies carry a measurable efficiency cost: EMVLight's own evaluation shows that traditional preemption techniques increase average non-EMV travel time by approximately 10% relative to unpreempted baselines [31]. These results indicate that emergency vehicle disruption is a first-order cost of operating a real traffic network, not a negligible edge case that a closed-world controller can safely ignore. Motivated by this, our framework treats EMVs and accident events as first-class constraints on the control problem rather than external events layered on top of it, at the cost of an increase in average travel time and delay relative to a disruption-free benchmark—a cost we treat as the price of a tighter approximation to real-world operating conditions.

Emergency vehicle routing and traffic signal pre-emption have also received considerable research attention. Conventional emergency routing methods generally employ shortest-path algorithms such as Dijkstra's algorithm, A*, and various time-dependent shortest path formulations to determine optimal routes under static or partially dynamic traffic conditions [25, 26, 27, 28, 29]. More recent studies have introduced reinforcement learning-based routing strategies capable of adapting to changing traffic environments [30]. EMVLight jointly optimizes emergency vehicle routing and adaptive traffic signal control using multi-agent actor–critic reinforcement learning, significantly reducing emergency response times while maintaining acceptable traffic efficiency for ordinary vehicles [31]. Subsequently, hierarchical multi-agent actor–critic methods have been proposed to further improve emergency routing through coordinated traffic signal control [33]. However, existing emergency traffic management approaches generally assume decentralized coordination and do not explicitly consider fairness, topology-aware regional decomposition, or simultaneous optimization for multiple emergency vehicles operating within large urban transportation networks. Visual sensing-based approaches have further demonstrated that incorporating real-time distance measurement and vehicle counting into traffic management can substantially improve EMV scheduling and intersection clearance [35].

Fairness has recently emerged as an important objective in intelligent transportation systems. FairMARL introduced a fairness-aware multi-agent reinforcement learning framework based on generalized social welfare functions to reduce disparities among intersections while maintaining competitive traffic efficiency [34]. Although fairness-aware optimization improves equitable resource allocation, existing approaches primarily focus on balancing intersection delays and do not integrate emergency vehicle routing, hierarchical coordination, or topology-aware traffic management into a unified framework.

Overall, significant progress has been achieved in analytical optimization, deep reinforcement learning, graph-based coordination, emergency vehicle routing, and fairness-aware traffic signal control. However, these objectives have largely been investigated independently. Classical optimization methods lack adaptability, deep reinforcement learning approaches prioritize traffic efficiency over fairness, graph-based coordination methods do not explicitly support emergency response, emergency vehicle routing methods overlook hierarchical city-wide coordination, and fairness-aware methods ignore emergency traffic management.

To the best of our knowledge, no existing framework combines hierarchical multi-agent reinforcement learning with automatically-compared, graph-based regional partitioning and an explicit ordinary-traffic fairness penalty within a single city-wide architecture. FairLight, presented in Section 3, implements this combination; city-wide multi-emergency-vehicle routing and a formal envy-free fairness guarantee are identified as open extensions rather than

Table 1: Comparison of FairLight against the closest related systems. "Not established" indicates the capability could not be verified from the cited source and should not be read as a confirmed absence.

Method	Hierarchical control	Automatic regionalization	Explicit EMV routing	Simultaneous multi-EMV	Ordinary-traffic fairness	Formal envy-free
CoLight [17]	No	No	No	No	No	No
MPLight [18]	No	No	No	No	No	No
HiLight [24]	Yes	No (fixed)	No	No	No	No
EMVLight [31]	No	No	Yes	Not established	No	No
FairMARL [34]	Not established	Not established	No	No	Yes	Not established
FairLight (this work)	Yes	Yes (compared)	Proposed	Proposed	Yes	Not implemented

claimed as implemented contributions (Section 8).

Table 1 summarizes how FairLight's verified and proposed components relate to the closest prior systems discussed above. Entries are marked "Yes" only where the cited paper or the FairLight implementation audited in this work explicitly supports the capability, "No" where the source explicitly does not, and "Not established" where the manuscript cannot verify the capability one way or the other from the cited source alone.

3. Proposed Framework

The proposed framework, FairLight, is organized around three components, two of which are implemented and evaluated in this paper and one of which is a design proposal. First, the road network is automatically partitioned into coherent subregions using graph-based community detection and density clustering (Louvain, Leiden, DBSCAN, and four hybrid combinations), removing the need for manual, expert-driven region design; this component is implemented and its effect on the resulting region structure is measured across five networks in Section 6. Second, the Sub-Policy reward is augmented with an explicit fairness penalty so that waiting-time disparity, starvation, and Jain's Fairness Index are optimized alongside local efficiency rather than left as an unmeasured side-effect; this component is implemented (Section 4) with closed-loop training evaluation pending. Third, as a design proposal rather than an implemented mechanism, the framework outlines how single-vehicle emergency routing could be extended to a multi-EMV setting in which several emergency vehicles simultaneously request priority across overlapping parts of the network, together with a mechanism for incident-robust training; neither is implemented in the current codebase, and none of the five evaluation scenarios currently contain emergency-vehicle demand. The remainder of this section describes each of these three components in turn, with implementation status stated explicitly at the start of each subsection.

3.1. Clustering Based on Louvain

Implementation status: implemented and evaluated (Section 6).

In the proposed framework, manually dividing the road network into precise subsections is impractical for real-world scenarios due to the large number of roads and intersections involved. Therefore, the road network is modeled as a weighted graph $G = (V, E)$, where intersections are represented as nodes and road segments as edges.

To automatically identify suitable subsections, the Louvain community detection algorithm (via the `python-louvain` implementation) is applied to the graph. The algorithm partitions the network into communities by maximizing modularity, which measures the difference between the observed connectivity within communities and the connectivity expected in a corresponding random network:

$$Q = \frac{1}{2m} \sum_{i,j} \left(A_{ij} - \frac{k_i k_j}{2m} \right) \delta(c_i, c_j), \tag{1}$$

where A_{ij} is the edge weight between nodes i and j, k_i and k_j are their weighted degrees, m is the total edge weight, and $\delta(c_i, c_j)$ indicates whether the nodes belong to the same community.

A higher modularity indicates stronger community structure, allowing the identified communities to serve as automatically generated road-network subsections. Each subsection is assigned a sub-policy, while a higher-level

meta-policy coordinates the behavior of these sub-policies across the complete network. This provides a scalable alternative to manual decomposition and enables hierarchical policy optimization over complex road networks.

3.2. Multiple EMV Vehicle Routing Across the City

Implementation status: proposed design, not implemented. No routing, conflict-arbitration, or priority-class code exists in the current codebase, and none of the five evaluation scenarios (Section 6) include an emergency vehicle in their demand data. The remainder of this subsection should be read as a design sketch for future work, not as a description of an evaluated system.

Prior emergency-vehicle (EMV) prioritization approaches in the traffic-signal-control literature, including the single-agent preemption strategies discussed in Section 2, are almost exclusively formulated around a single emergency vehicle traversing the network at any given time. This assumption breaks down in dense urban environments, where it is common for multiple ambulances, fire trucks, and police vehicles to be simultaneously active, often converging on overlapping segments of the road network from different origins and with different destinations. Extending EMV prioritization to a genuinely multi-vehicle setting therefore requires the framework to reason jointly about several time-critical routes rather than optimizing each one in isolation.

We propose to model the multi-EMV routing problem as a joint optimization over a set of active emergency vehicles $\mathcal{E} = \{e_1, e_2, \ldots, e_K\}$, each associated with a current position, a destination, and a priority class (for example, life-threatening medical emergency versus structural fire response). At each decision step, the framework must determine (i) a preferred route for every active emergency vehicle through the signalized network, and (ii) how to allocate green-phase priority at any intersection that lies on more than one emergency vehicle's preferred route. This second requirement is the central new difficulty relative to single-EMV routing: when two emergency vehicles approach the same intersection from conflicting directions within a short time window, granting unconditional priority to both is physically impossible, and the controller must instead arbitrate based on factors such as estimated time-to-intersection, remaining distance to destination, and priority class.

We envision extending the existing Meta-Policy/Sub-Policy hierarchy so that each active emergency vehicle contributes an additional, time-decaying bias term to the regional and intersection-level state representations along its anticipated route, rather than introducing a wholly separate routing module. Conceptually, this bias term would increase in magnitude as an emergency vehicle approaches a given intersection and decay once it has passed, so that the learned Sub-Policy naturally favors clearing a path ahead of an approaching emergency vehicle while reverting to its ordinary efficiency/fairness trade-off once the vehicle has moved on. Coordination across subregions, needed when an emergency vehicle's route crosses several Louvain-derived subregions in sequence, would be handled by the Meta-Policy, which is already responsible for issuing cross-regional guidance.

Resolving conflicts between multiple simultaneously active emergency vehicles is expected to require an explicit arbitration rule layered on top of this bias mechanism — for instance, prioritizing the vehicle with the least estimated time-to-intersection, breaking ties by priority class, and bounding the maximum delay that can be imposed on a lower-priority emergency vehicle so that no single vehicle is perpetually deprioritized in favor of others. The corresponding evaluation methodology would extend the single-EMV response-time metrics used elsewhere in this work (e.g., average and 95th-percentile emergency waiting time) to the multi-vehicle case by additionally reporting the spread of response-time improvements across all active emergency vehicles, so that the framework can be shown to accelerate every emergency vehicle's passage and not merely the one considered by the controller to be of highest priority.

Finally, we note that a multi-EMV setting also changes how the framework's fairness objective (Section 3.3) should be interpreted: granting priority to emergency vehicles necessarily imposes additional waiting time on ordinary traffic, and this additional cost must be distributed across the network rather than concentrated on whichever ordinary vehicles happen to be sharing a road segment with an active emergency-vehicle corridor at a given moment. This interaction between multi-EMV routing and the fairness reward is left as an open design question to be resolved in the full version of this section.

3.3. Fairness and Accidental Congestion Control

Implementation status: the fairness penalty described below (variance, starvation threshold, Jain's Fairness Index) is implemented and used in training (Section 4); the envy term and the accident/incident-injection mechanism described later in this subsection are proposed designs, not implemented in the current codebase.

Traffic-signal controllers optimized purely for network-wide efficiency metrics, such as average travel time or average delay, can produce policies that are efficient in aggregate yet highly inequitable at the level of individual approaches or vehicles. A controller may, for instance, learn to sacrifice one chronically low-priority approach in order to keep throughput high on the network's dominant corridors, resulting in a small subset of vehicles experiencing persistent, disproportionate waiting times even while the network-wide average delay looks acceptable. The fairness component of the proposed framework is intended to make this trade-off explicit and controllable rather than an incidental by-product of reward design.

We plan to quantify fairness along several complementary axes: the variance of waiting times across approaches at a given intersection, the maximum (tail) waiting time experienced by any single approach, and Jain's Fairness Index computed over per-approach or per-vehicle service levels. Unlike a single scalar efficiency term, these measures are each sensitive to a different failure mode — variance captures persistent imbalance between approaches, the tail-waiting-time penalty specifically targets starvation of the worst-served approach, and Jain's index provides a normalized, scale-independent summary of overall equity that can be tracked and compared across scenarios and partitioning methods. An "envy" term, comparing each intersection's service level against that of its neighboring intersections, is further intended to discourage locally greedy policies that improve one intersection's performance strictly at a neighbor's expense — directly relevant to the Louvain-based regional decomposition of Section 3.1, since poorly balanced subregions could otherwise induce exactly this kind of neighbor-level inequity.

A second, largely orthogonal concern addressed by this subsection is robustness to accidental congestion: collisions, breakdowns, and other unplanned incidents that suddenly and sharply reduce the effective capacity of one or more road segments. Conventional traffic signal controllers are typically trained and evaluated exclusively under nominal demand conditions, and so may respond poorly when a lane or approach is unexpectedly blocked, continuing to allocate green time as though the obstructed segment retained its usual capacity. We intend to address this by exposing the controller to a representative distribution of injected incidents during training — varying in location, duration, and severity — so that the learned Sub-Policy and Meta-Policy learn to detect the resulting shift in local traffic statistics (e.g., a sudden rise in queue length or a collapse in mean speed on an otherwise normal approach) and reallocate green time accordingly, rather than requiring a separate, hand-engineered incident-detection and override layer.

We additionally plan to evaluate whether accident-aware training degrades the framework's fairness guarantees: an incident response that simply redirects all available green time toward the obstructed approach could, in principle, starve unrelated approaches elsewhere in the network. The evaluation methodology for this subsection will therefore report fairness metrics (variance of waiting time, tail waiting time, Jain's index) separately for incident and incident-free simulation intervals, to confirm that the controller's response to accidental congestion does not come at the cost of the equity objectives established earlier in this section. Together with the multi-EMV routing extension of Section 3.2, this completes the three methods — scalable graph-based regional decomposition, coordinated multi-emergency-vehicle routing, and fairness-aware, incident-robust congestion control — that constitute the proposed FairLight framework.

3.4. Hierarchical Architecture Overview

Implementation status: unless otherwise noted, the architecture described in Sections 3.4–3.10 (Meta-Policy, Sub-Policy, PPO training, and safety-constrained execution) is implemented and verified against the training code; closed-loop evaluation results are reported separately in Section 6 and are pending at the time of writing.

The proposed FairLight framework addresses the scalability and coordination challenges of city-scale adaptive traffic signal control through a hierarchical multi-agent reinforcement learning architecture. A fully centralized controller becomes computationally expensive as the number of intersections increases, whereas independent intersection-level agents lack sufficient network-wide context to coordinate traffic across connected corridors. To address this trade-off, FairLight introduces an intermediate regional abstraction between network-level coordination and local signal control.

The framework consists of four principal components: (i) automatic regional construction, (ii) a network-level Meta-Policy, (iii) intersection-level Sub-Policies, and (iv) a safety-constrained signal execution layer. Regional construction transforms the signalized road network into coherent traffic regions, the Meta-Policy extracts global spatial-temporal information and generates network-level control targets, and the Sub-Policies translate this global guidance into local phase decisions using intersection and neighborhood information. The resulting hierarchy is trained jointly using PPO with an adversarial goal-alignment mechanism.

3.5. Traffic Network Representation and Automatic Regionalization

Implementation status: implemented and evaluated (Section 6). Region membership is computed once, offline, before training begins, and is loaded from a static cluster file for the entire training and evaluation run; only the edge weights feeding Louvain and Leiden (below) are estimated from a traffic simulation pass, not the community structure itself. This is a dynamic-weight, static-membership design: it should not be described as continuously adaptive regionalization.

Manually decomposing a large urban road network into meaningful control regions is impractical because the number and connectivity of signalized intersections can vary substantially across networks. FairLight therefore represents the signalized road network as a weighted graph

$$G = (V, E), \tag{2}$$

where each vertex $v_i \in V$ represents a signalized intersection and each edge $e_{ij} \in E$ represents the connectivity between two signalized intersections. Non-signalized junctions between traffic lights are contracted during graph construction so that the resulting graph directly represents the controllable signalized-intersection topology.

The connectivity between intersections is characterized using a traffic-dependent edge weight rather than purely static topology. For an edge e, the implemented weighting function is

$$w_e = \alpha n_e + \beta \omega_e + \gamma \kappa_e n_e, \tag{3}$$

where n_e denotes the number of vehicles on the edge, ω_e represents accumulated waiting time, and

$$\kappa_e = \mathrm{clip}\left(1 - \frac{v_e}{v_e^{\max}}, 0, 1\right) \tag{4}$$

denotes the congestion ratio. The implementation uses $\alpha = 1.0, \beta = 0.3$, and $\gamma = 2.0$. These weights are averaged over a 1000-step simulation window, allowing the regional structure to reflect persistent traffic interactions rather than a transient traffic state.

The resulting weighted graph is partitioned into communities using graph community-detection methods. In particular, Louvain and Leiden identify communities by maximizing graph modularity,

$$Q = \frac{1}{2m} \sum_{i,j} \left(A_{ij} - \frac{k_i k_j}{2m}\right) \delta(c_i, c_j), \tag{5}$$

where A_{ij} is the weighted adjacency matrix, k_i and k_j are the weighted degrees of vertices i and j, m is the total edge weight, and $\delta(c_i, c_j)$ indicates whether the two vertices belong to the same community. Louvain is implemented via the `python-louvain` package and Leiden via `leidenalg` with the `ModularityVertexPartition` objective, both operating on the traffic-weighted graph above; both post-process their raw output by merging singleton or near-singleton communities into a neighboring community before the partition is written out, so the reported region counts in Section 6 already reflect this merge step.

DBSCAN is implemented as a separate, spatial clustering path: it clusters intersections by their (x, y) coordinates under Euclidean distance, not by the traffic-weighted graph used for Louvain and Leiden. Its `eps` and `min_samples` parameters are auto-tuned rather than fixed: `min_samples` is set to the network's node count divided by 10, clipped to $[3, 8]$, and `eps` is swept over a set of log-spaced candidates derived from k-nearest-neighbor distances, selecting the candidate with the lowest noise-point fraction whose resulting mean cluster size falls in an 8–40-node target band. The four hybrid methods combine the two differently depending on composition order: `dbscan_X` (for community-detection method X) first applies spatial DBSCAN to form fine-grained spatial cells and then runs X on the resulting cell-adjacency supernode graph, while `X_dbscan` first runs community detection X on the traffic-weighted graph and then subdivides each resulting community spatially using DBSCAN. Because DBSCAN's clustering criterion is purely spatial, it does not directly incorporate the traffic-dependent edge weight w_e defined above; only the Louvain, Leiden, and hybrid paths that invoke community detection do so.

This regionalization produces a set of regions

$$\mathcal{R} = \{R_1, R_2, \ldots, R_M\}, \tag{6}$$

which forms the intermediate abstraction used by the hierarchical controller. Each region provides aggregated traffic information to the Meta-Policy, while individual intersections within the regions remain independently controllable by the Sub-Policies.

3.6. Hierarchical Meta-Policy

The Meta-Policy provides network-level coordination by observing the temporal evolution of traffic conditions across the generated regions. For each region R_k, the regional state is represented as

$$s_k^t = \left[N_k^t, W_k^t, x_k, y_k\right] \in \mathbb{R}^4, \tag{7}$$

where N_k^t denotes the aggregate stopped-vehicle count, W_k^t denotes aggregate waiting time, and (x_k, y_k) represents the normalized spatial centroid of the region.

Because traffic conditions are inherently temporal, the Meta-Policy maintains a sliding window of $T = 20$ regional observations. A Transformer encoder processes the regional tokens jointly to capture spatial dependencies between regions. A learnable global token is introduced to obtain a network-level representation

$$F_g \in \mathbb{R}^4. \tag{8}$$

The temporal representations of individual regions are subsequently processed using an LSTM. The resulting hidden states are concatenated across the M regions and transformed through an MLP to obtain a global representation $G \in \mathbb{R}^{16}$. The Meta-Policy then generates two network-level control targets,

$$(G_w, G_q) = \mathrm{Linear}_{16\to 2}(G), \tag{9}$$

where G_w represents the desired global waiting-time level and G_q represents the desired global queue-length level.

Thus, rather than directly selecting signal phases throughout the network, the Meta-Policy establishes a global traffic-control objective that can be pursued by the lower-level intersection agents. This separation allows long-range traffic coordination to be handled at a compressed regional level while preserving local responsiveness at individual intersections.

3.7. Intersection-Level Sub-Policy

Implementation status: implemented. The neighborhood mechanism described below is a fixed-capacity, top-4 attention layer, not an arbitrary-degree mechanism; see the clarification after Eq. (12).

The Sub-Policy converts global guidance into local signal-control actions. At each intersection i, the local observation is represented by

$$o_i^t = \left[v_i, q_i, o_i, f_i, s_i, w_i, v_i^{avg}, p_i, c_i, d_i\right] \in \mathbb{R}^{10}, \tag{10}$$

where the components capture local traffic volume, queue length, occupancy, flow, stops, waiting time, speed, pressure, congestion, and delay. The observations are normalized using an exponential moving average and clipped to a bounded range before being supplied to the policy network.

A local multilayer perceptron first encodes the intersection state. Since traffic conditions at one intersection are influenced by neighboring intersections, the encoded representation is augmented using a Graph Attention Concat (GAC) mechanism over the intersection-adjacency graph. For a neighboring pair (i, j), the attention coefficient is computed as

$$e_{ij} = \mathrm{LeakyReLU}\Big(a^T[h_i \| h_j]\Big), \qquad \alpha_{ij} = \mathrm{softmax}_j(e_{ij}), \tag{11}$$

and the resulting neighborhood representation is constructed through concatenation,

$$z_i = h_i \| \alpha_{i1} h_{j_1} \| \cdots \| \alpha_{i4} h_{j_4}. \tag{12}$$

The neighbor set $\{j_1, \ldots, j_4\}$ in Eq. (12) is a fixed-capacity slot of exactly four, not a variable-length representation of an intersection's true adjacency degree: when an intersection has more than four signalized neighbors, the remainder

are truncated from the representation; when it has fewer than four, the unused slots are zero-padded. The regional partitioning of Section 3.5 determines *which* intersections are grouped into a region, but the attention mechanism itself operates on this fixed four-neighbor window regardless of an intersection's actual degree in the signalized road network. Characterizing this mechanism as adapting to arbitrary network degree would overstate what is implemented; its correct characterization is a capped, zero-padded neighborhood of size four.

The neighborhood representation is then combined with the global Meta-Policy embedding:

$$s_i^t = \left[z_i \| F_g \right]. \tag{13}$$

This representation simultaneously provides the Sub-Policy with local traffic information, neighboring traffic context, and network-level coordination guidance.

A shared actor-critic network processes this representation and produces a probability distribution over signal phases. Because intersections may have different numbers of valid signal phases, invalid action indices are masked before the final softmax operation. Parameter sharing across intersections allows the same learned policy structure to be deployed across heterogeneous signalized intersections while retaining local observations and graph-based neighborhood information.

3.8. Joint Hierarchical Optimization

The Meta-Policy and Sub-Policies are optimized jointly. The Sub-Policy is trained using Proximal Policy Optimization (PPO) with Generalized Advantage Estimation (GAE). The temporal-difference residual is

$$\delta_t = r_t + \gamma V(s_{t+1})(1 - d_t) - V(s_t), \tag{14}$$

and the generalized advantage estimate is

$$A_t = \delta_t + \gamma\lambda(1 - d_t)A_{t+1}, \tag{15}$$

with $\gamma = 0.99$ and $\lambda = 0.95$. The PPO policy objective uses the clipped probability ratio

$$L_{\text{policy}} = -\min\left(r_t A_t, \text{clip}(r_t, 1 - \epsilon, 1 + \epsilon)A_t\right), \tag{16}$$

where $\epsilon = 0.2$.

The hierarchical relationship is enforced through an additional goal-alignment term. Let W_{global} and Q_{global} denote the actual network-wide waiting time and queue length. The Meta-Policy is trained using

$$L_{\text{Meta}} = \text{MSE}\Big((G_w, G_q), (W_{\text{global}}, Q_{\text{global}})\Big) + \eta_1 r_g, \tag{17}$$

while the Sub-Policy receives an alignment objective encouraging the realized network state to follow the Meta-Policy's targets:

$$L_{\text{Sub}} = L_{\text{AC}} + \eta_2 \left[\beta_q(Q_{\text{global}} - G_q) + \beta_w(W_{\text{global}} - G_w)\right]. \tag{18}$$

Here, $\eta_1 = \eta_2 = 0.1$ and $\beta_q = \beta_w = 0.5$. The alignment gradient is detached from the Meta-Policy in the Sub-Policy loss, preventing the lower-level policy from directly modifying the higher-level representation through this term.

The local reward combines intersection-level traffic efficiency with the global goal-alignment reward:

$$r_i^t = -\left(q_i^t + w_i^t + d_i^t + p_i^t - s_i^t\right), \tag{19}$$

$$r_g^t = -\left[\beta_q(W_{\text{global}} - G_q) + \beta_w(Q_{\text{global}} - G_w)\right], \tag{20}$$

and

$$r^t = r_i^t + r_g^t. \tag{21}$$

This formulation ensures that each intersection is evaluated according to its own local traffic conditions while simultaneously receiving a network-level coordination signal.

3.9. Safety-Constrained Signal Execution

The learned policy does not directly bypass traffic-signal safety constraints. Every requested phase transition is passed through a deterministic execution layer that enforces a minimum green interval followed by a yellow transition before the new green phase is activated.

Specifically, a requested phase change is permitted only after the current phase has satisfied the minimum-green requirement. The controller then enters a yellow transition for the prescribed duration before committing the requested green phase. In the current implementation, the minimum green time is 5 simulated seconds and the yellow transition lasts 3 seconds.

This separation between learned decision-making and deterministic safety enforcement prevents the reinforcement-learning agent from producing physically invalid or unrealistically rapid signal transitions.

3.10. End-to-End Control Process

At every reinforcement-learning decision step, the framework first aggregates the current traffic state for each automatically generated region and updates the temporal regional-state buffer. The Meta-Policy then processes this temporal representation to obtain the global embedding and network-level queue and waiting-time targets. Simultaneously, each intersection obtains its normalized local observation, incorporates information from neighboring intersections through graph attention, and fuses this representation with the global Meta-Policy embedding.

The resulting Sub-Policy actions are passed through the safety-constrained signal execution layer before being applied to the traffic simulator. The environment then advances the traffic simulation, computes local and global rewards, and stores the resulting transition for policy optimization. Meta-Policy updates are performed periodically, while Sub-Policy PPO updates are performed after the rollout buffer reaches the specified horizon.

The complete architecture can therefore be expressed as the following hierarchical mapping:

$$\begin{aligned} G \rightarrow \mathcal{R} \rightarrow \text{Meta-Policy} \rightarrow F_g, (G_w, G_q) \\ \rightarrow \text{Sub-Policies} \rightarrow \text{Signal Actions} \rightarrow \text{Traffic Environment}. \end{aligned} \tag{22}$$

This design separates regional coordination, temporal network reasoning, local spatial interaction, and physical signal execution into distinct layers. Consequently, the framework can automatically adapt its regional abstraction to different road networks while retaining a shared local control mechanism and a global coordination signal.

4. Multi-Objective Reward Function Design

An effective traffic signal controller must optimize multiple objectives simultaneously rather than maximizing traffic throughput alone. Conventional reinforcement learning approaches primarily optimize queue lengths or average waiting times, often producing policies that systematically disadvantage certain intersections and fail to prioritize emergency vehicles. This section first states the design target for a complete multi-objective reward, then states precisely which part of that design target is implemented in the current training code, to avoid the two being conflated.

Design target. The intended overall reward at each decision step is

$$R_{\text{total}} = w_1 R_{\text{eff}} + w_2 R_{\text{fair}} + w_3 R_{\text{emergency}}, \tag{23}$$

with $w_1 + w_2 + w_3 = 1$ and $w_1 \geq w_2 \geq w_3 \geq 0$, assigning R_{eff} the largest weight because it governs global traffic-congestion control across the whole network rather than emergency-vehicle service alone, followed by fairness and then emergency responsiveness.

As implemented. The weighted combination above, with explicit w_1, w_2, w_3 coefficients, is *not* present in the training code: no such weight variables exist, and $R_{\text{emergency}}$ is not added to the reward the Sub-Policy is trained on. What *is* implemented and used for training is the local reward of Section 3.8, $r_i^t = -(q_i^t + w_i^t + d_i^t + p_i^t - s_i^t)$, an absolute (not temporal-difference) combination of queue length, waiting time, delay, pressure, and speed, plus the fairness penalty $R_{\text{fair},j}$ defined later in this section, plus the goal-alignment term r_g^t. The discussion below presents R_{eff} and $R_{\text{emergency}}$ as proposed refinements to be reconciled with r_i^t in future work, not as a description of the currently trained objective.

4.1. Traffic Efficiency Reward (proposed refinement)

Implementation status: not implemented as stated below; the efficiency term actually used in training is r_i^t from Section 3.8. The following formulation is proposed as a denser-signal refinement of r_i^t, encouraging continuous improvement in network performance by rewarding reductions in congestion rather than rewarding only low congestion states:

$$R_{\text{eff}} = -\beta_1 \Delta Q - \beta_2 \Delta W + L_{\text{avg}}, \tag{24}$$

where ΔQ denotes the change in total queue length between successive decision steps, ΔW represents the change in cumulative vehicle waiting time, and L_{avg} is the average local throughput reward aggregated across all intersections. The coefficients β_1 and β_2 determine the relative importance assigned to queue reduction and waiting-time reduction.

Unlike conventional reward formulations based solely on absolute queue lengths, the use of temporal differences would in principle provide denser learning signals by rewarding measurable improvements between consecutive time steps; whether this translates into faster convergence in practice has not been tested, since this term is not yet wired into the training loop.

4.2. Fairness Reward

Implementation status: implemented. The coefficients and thresholds below are literal values in the training code (`environment.py`), verified against the implementation.

Conventional traffic-signal controllers commonly optimize aggregate measures such as queue length, average delay, and throughput. These objectives can improve network efficiency but do not guarantee equitable service to individual vehicles: a movement repeatedly deprioritized by the controller may experience prolonged waiting even when network-level metrics improve. FairLight therefore treats fairness as a complement to traffic efficiency and emergency-vehicle responsiveness, consistent with the multi-objective reward defined above.

Vehicle waiting-time monitoring.. For each vehicle v and intersection j, let $w_{v,j}(t)$ denote its accumulated waiting time at time t. The timer increases by one simulation step whenever the vehicle is stopped (speed below 0.1 m/s) on an incoming lane of j, is kept while the vehicle creeps forward in the queue, and is discarded when the vehicle is served and leaves the incoming lanes of j; a new waiting episode then starts at the next intersection. The same definition is used in training and evaluation. A vehicle is flagged as a starvation risk when

$$w_{v,j}(t) \geq W_{\max}, \qquad W_{\max} = 90 \text{ s}. \tag{25}$$

Reaching $W_{\max}$ flags the vehicle and increases its consideration in subsequent phase selection: the requested phase of the Sub-Policy is replaced by a feasible phase that gives a green to the lane of the longest-waiting flagged vehicle (the current phase is kept if it already does). The 90-second threshold is a design threshold and not a universal guarantee that every vehicle is served within 90 seconds: demand, conflicting movements, incidents, clearance intervals, and physical feasibility may make the threshold temporarily or persistently unattainable. The override only changes the *requested* phase; the controller never bypasses the minimum-green requirement (5 steps) or the 3 s yellow clearance of Section 3.9, and no override is issued during a yellow transition.

To align this safeguard with the reward, define the local maximum waiting time at intersection j as $W_j(t) = \max_{v \in \mathcal{V}_j(t)} w_{v,j}(t)$, where $\mathcal{V}_j(t)$ is the set of tracked vehicles currently waiting at that intersection. The threshold penalty is then $\max(0,\, W_j(t) - W_{\max})$. The maximum is taken locally per intersection, not over the whole episode or network.

Intersection-level fairness using Jain's index.. At each intersection, Jain's Fairness Index (JFI) summarizes how evenly a set of non-negative service scores is distributed. For n included vehicles with scores $x_i \geq 0$,

$$J = \frac{\left(\sum_{i=1}^{n} x_i\right)^2}{n \sum_{i=1}^{n} x_i^2}. \tag{26}$$

For non-zero scores, $J \in [1/n, 1]$, and a value closer to 1 indicates a more even distribution. If all scores are zero the denominator vanishes; we adopt the convention $J = 1$ (no disparity) and, when no vehicle is waiting at all, set every fairness term to zero.

The choice of x_i matters. Raw waiting time should not be used as a service score, because a higher waiting time indicates *worse* service, and equal but very high waiting times still yield $J = 1$. We therefore use the bounded, non-negative score that is inversely related to waiting time,

$$x_i = \frac{1}{1 + w_{i,j}/W_{\text{ref}}}, \tag{27}$$

where $W_{\text{ref}} = 30$ s is a fixed reference time chosen before training. The score, the population (all vehicles waiting on the incoming lanes of j), the sampling interval (every simulation step), and the handling of vehicles not yet served are identical across methods and experiments. JFI measures equality, not absolute performance or starvation by itself, so it is reported alongside the mean, maximum, and 95th-percentile waiting time, queue length, throughput, and emergency-vehicle response time, and it is computed locally per intersection before any city-wide aggregation so that local disparities are not concealed by a network average.

Fairness reward.. Let W_j be the maximum waiting time at intersection j as defined above, let $\sigma^2_{\hat{w},j}$ be the variance of the normalized waiting times $\hat{w}_i = w_{i,j}/W_{\text{ref}}$ over the local population, and let J_j be the JFI of the scores x_i. The dimensionless local fairness penalty is

$$P_{\text{fair},j} = \lambda_{\text{var}} \operatorname{Var}(\hat{w}) + \lambda_{\text{starve}} \max\Big(0, \frac{W_j - W_{\max}}{W_{\max}}\Big) + \lambda_J (1 - J_j), \tag{28}$$

and the fairness reward is $R_{\text{fair},j} = -P_{\text{fair},j}$, which is added to the local reward r^t_j of Section 3.8. Normalizing waiting times by W_{ref} and the starvation excess by $W_{\max}$ is important because waiting-time variance has squared-time units whereas the starvation and JFI terms are dimensionless. The coefficients $\lambda_{\text{var}}, \lambda_{\text{starve}}, \lambda_J \geq 0$ are hyperparameters; larger disparity, threshold excess, or inequality reduces the reward. Their initial values $(\lambda_{\text{var}}, \lambda_{\text{starve}}, \lambda_J) = (1, 10, 10)$ were chosen so that the fairness terms are not negligible next to the queue and waiting terms of r^t_j; they are design choices, not empirically established optimal values. The earlier coefficients 0.3, 0.25 and 0.15 applied to quantities with incompatible units and are superseded by this formulation. Envy and pedestrian terms are not part of the implemented fairness reward.

4.3. Emergency Vehicle Reward (proposed, not wired into training)

Implementation status: not implemented. The environment code includes a read-only helper that computes cumulative emergency-vehicle waiting time for monitoring purposes, but this quantity is never added to the reward the policy is trained on, and no scenario's current demand data contains an emergency vehicle for it to measure. The formulation below is therefore a proposed reward term, not a description of current training behavior.

Emergency vehicles would require substantially higher priority than regular traffic because delays directly affect emergency response times. The proposed reward component is

$$R_{\text{emergency}} = -W_{\text{emergency}}, \tag{29}$$

where $W_{\text{emergency}}$ is the cumulative waiting time experienced by emergency vehicles. This formulation would directly penalize any delay encountered by ambulances, fire trucks, or police vehicles, encouraging the controller to allocate green phases immediately upon emergency vehicle detection, once (i) emergency vehicles are added to the simulated demand and (ii) this term is connected to the training loop.

Finally, the combined reward is normalized using an exponential moving average before being supplied to the reinforcement learning algorithm. Reward normalization reduces reward variance, improves numerical stability during training, and accelerates policy convergence.

4.4. Evaluation Protocol

Fairness is evaluated jointly with efficiency and safety. At minimum we report the mean, maximum, and 95th-percentile vehicle waiting time; the proportion of vehicles exceeding 90 seconds; intersection-level JFI; queue length; throughput; and emergency response time. The controller is compared against the same controller without fairness terms, under identical traffic demand, random seeds where applicable, signal constraints, and simulation horizons,

and multiple runs are reported with variability estimates. A reduction in average delay alone is insufficient evidence of improved fairness, and a higher JFI alone is insufficient unless absolute delays remain acceptable. Emergency pre-emption intentionally gives some vehicles higher urgency, so results distinguish ordinary-vehicle fairness from emergency-vehicle responsiveness. At present, the training and evaluation code does not set a random seed for either the simulator or the learning algorithm (no seeding call exists in the training entry point), so repeated runs under this protocol are not currently reproducible run-to-run; adding explicit seed control is a prerequisite for the multi-run variability estimates this protocol calls for, and is tracked as a limitation in Section 7.

5. Experimentation

5.1. Simulation Environment

The proposed framework is evaluated in the Simulation of Urban MObility (SUMO) environment via the TraCI interface. SUMO is chosen because its microscopic, lane-level vehicle dynamics and configurable signal infrastructure allow the simulated network to closely approximate the congestion and coordination patterns of a real signalized road network, providing a controlled setting in which the proposed regional partitioning, fairness, and emergency-routing mechanisms can be exercised repeatedly under known traffic demand. A simulation study, however, remains an approximation: real-world deployment would need to contend with sensor noise, imperfect vehicle detection, driver non-compliance, and infrastructure constraints that are not fully captured by SUMO's vehicle models. Validating the framework on a real traffic signal system is therefore treated as a necessary step beyond the present simulation-based evaluation, and not as a substitute for it.

5.2. Base Model and Benchmark Methods

The base model used throughout this evaluation is the Hierarchical Region-Aware Multi-Agent Reinforcement Learning controller with the fairness reward of Section 3.3 enabled (Meta-Policy + Sub-Policy, trained with the multi-objective reward of Section 4). This base model is compared against two traditional, non-learning traffic signal control strategies commonly used as references in the traffic-signal-control literature:

- *Fixed Time*: a cyclical, predefined signal-phase schedule with a fixed green/yellow split and no responsiveness to real-time traffic conditions.
- *Max Pressure*: a reactive, non-learning controller that selects the signal phase maximizing the pressure difference between incoming and outgoing lanes at each intersection [5].

The city-wide multi-EMV routing comparison described in some of this framework's design documents is *not* currently executable: the routing and arbitration mechanism of Section 3.2 is unimplemented, and none of the five scenarios below include an emergency vehicle in their demand data, so there is no EMV traffic for an "with vs. without city-wide routing" comparison to act on. Where such a comparison appears in Section 6 it is explicitly marked as blocked on these two prerequisites, not merely pending compute time.

5.3. Traffic Network Scenarios and Experimental Configuration

The framework is configured for five SUMO networks, selected to vary network scale, topology regularity, and provenance (real-world OSM extracts versus synthetic grids). Table 2 reports the verified configuration of each, obtained directly from the `.net.xml` and demand files shipped in the repository; entries are marked "Not reported" where the corresponding configuration file does not specify a value, rather than inferring one.

Arterial4x4 and Grid4x4 each ship 1400 pre-generated, numbered demand variants in addition to the default (`_1`) file reported above; these appear intended as seed/demand variants for repeated runs but are not yet invoked by any automated sweep in the training or evaluation code, so the single-run vehicle counts in Table 2 reflect only the default configuration. EMV origin, destination, and priority-class fields, and the number of regional communities generated per scenario, are reported in Section 6 (the latter) or marked not applicable (the former, since no scenario currently defines an EMV).

Table 2: Traffic network scenarios used for regionalization evaluation (Section 6). Intersection counts are signalized (traffic-light) junctions in the contracted intersection graph of Section 3.5; vehicle counts are the configured demand for the default route file. None of the five scenarios currently include an emergency vehicle in their demand data.

Scenario	Intersections	Edges	Vehicles	Sim. window [s]	Network type
Manhattan	472	6184	1800	Not reported	OSM, real-world
Cologne8	8	590	2046	25200–28800 (3600)	OSM, real-world
Ingolstadt21	21	2415	4283	57600–61200 (3600)	OSM, real-world
Arterial4x4	16	320	2484	0–3600 (3600)	Synthetic, arterial grid
Grid4x4	16	464	1473	0–3600 (3600)	Synthetic, grid

Table 3: Regions produced per scenario and method: region count n, mean region size (intersections/region), and modularity Q where applicable (DBSCAN does not optimize modularity, since it clusters on spatial coordinates rather than the traffic-weighted graph; Section 3.5). Values are the current repository's cluster output.

Scenario	DBSCAN (n, mean size)	Louvain (n, mean size, Q)	Leiden (n, mean size, Q)
Manhattan	8, 59.0	9, 52.4, 0.620	70, 6.7, 0.577
Cologne8	1, 8.0	1, 8.0, 0.000	2, 4.0, 0.012
Ingolstadt21	2, 10.5	1, 21.0, 0.005	2, 10.5, 0.052
Arterial4x4	1, 16.0	2, 8.0, 0.414	5, 3.2, 0.153
Grid4x4	1, 16.0	1, 16.0, 0.375	5, 3.2, 0.288

5.4. Neighborhood Construction

The Sub-Policy's Graph Attention Concat mechanism (Section 3.7, Eq. (12)) operates on a fixed-capacity window of the four highest-relevance signalized neighbors per intersection, with truncation above four neighbors and zero-padding below four. This is not a variable-degree or topology-adaptive mechanism: it is capped regardless of an intersection's true degree in the signalized road network. Whether any of the five scenarios in Table 2 contain signalized intersections with more than four neighbors — and therefore lose information to truncation — has not been audited at the time of writing; this is listed as an open question in Section 7.

5.5. Evaluation Metrics

Four categories of metric are used, following the structure of Section 6. Only metrics collectible from the existing TraCI instrumentation or from the static clustering output are reported as measured; the remainder require a completed training/evaluation run and are reported as pending.

Traffic efficiency: total average trip time (vehicles completing their trip within the simulation horizon), total waiting time, and total average waiting time, consistent with Section 3.8's reward terms.

Emergency responsiveness: mean, median, and 95th-percentile EMV response time, and route-completion rate — all currently inapplicable, since no scenario contains an EMV.

Ordinary-vehicle fairness: Jain's Fairness Index, mean and 95th-percentile waiting time, the proportion of vehicles exceeding the $W_{\max} = 90$ s threshold, and per-intersection waiting-time variance, as defined in Section 4.

Scalability and regionalization: number of detected regions, region-size distribution, and modularity per partitioning method. These are computed directly from the static clustering output and do not require a trained policy; they are reported as measured results in Section 6.

6. Results and Discussion

6.1. Regionalization Comparison (measured)

Unlike the tables later in this section, Table 3 reports measured output of the implemented clustering pipeline (Section 3.5) applied to all five scenarios of Table 2: no trained policy is required to produce these numbers, so they are reported as actual results rather than placeholders.

Table 4: Base model with vs. without city-wide EMV routing — placeholder structure; blocked on unimplemented routing code and the absence of EMV demand in all five scenarios (Table 2).

Configuration	Trip Time [s]	Waiting Time [s]	Avg. Waiting Time [s]
Without city-wide EMV routing	0.00	0.00	0.00
With city-wide EMV routing	0.00	0.00	0.00

Table 5: Comparison of traffic signal control methods — placeholder values pending a completed, seeded, multi-run evaluation campaign.

Method	Trip Time [s]	Waiting Time [s]	Avg. Waiting Time [s]
Fixed	0.00	0.00	0.00
Max Pressure	0.00	0.00	0.00
Algorithm	0.00	0.00	0.00

These results answer RQ1 (Section 1): the choice of regionalization method materially changes the structure handed to the Meta-Policy, and the effect is network-dependent rather than uniform. On Manhattan — the largest, most topologically rich network — Leiden produces far more, far smaller regions than Louvain (70 vs. 9), with 34 of Leiden's 70 regions being singletons despite the singleton-merging post-processing of Section 3.5; both achieve comparable, reasonably high modularity (0.577–0.620). On the three smaller or sparser networks (Cologne8, Ingolstadt21), Louvain and DBSCAN both frequently collapse to a single region covering the entire network, and modularity is close to zero (Cologne8 Louvain: Q=0) — indicating these networks lack strong community structure at this scale, so "automatic regionalization" degenerates to no regionalization at all for them under Louvain. DBSCAN, which clusters purely on intersection coordinates rather than traffic load, produces coarser and more size-uniform partitions across all five networks than either community-detection method, consistent with it optimizing a different (spatial) criterion rather than a worse one. These results indicate that the choice of regionalization method should be validated per-network rather than fixed globally, and that a method producing many small regions on one network (Leiden on Manhattan) can produce almost none on another (Leiden on Cologne8), which is important context for interpreting any downstream control-performance difference once closed-loop results are available.

6.2. Effect of City-Wide EMV Routing (blocked, not merely pending)

Table 4 is the comparison this framework would support between the base model operated with city-wide multi-EMV routing (Section 3.2) and the same base model without it. Unlike Table 5, this comparison is currently blocked on two unmet prerequisites rather than simply awaiting compute time: the routing and arbitration mechanism itself is unimplemented, and no scenario's demand data contains an emergency vehicle for the comparison to act on. The entries below are placeholders recorded to make the intended comparison explicit; they must not be read as measured results, and should not be populated until both prerequisites are resolved.

6.3. Comparison with Baseline Traffic Control Methods (pending)

Table 5 reports the three evaluation metrics of Section 5.5 for Fixed Time, Max Pressure, and the proposed base model (Algorithm). Unlike Section 6.2, this comparison is not blocked by missing functionality — Fixed Time, Max Pressure, and the PPO-trained base model are all implemented — but the corresponding simulation runs have not yet been executed under the current (PPO-based) training code at the time of writing, and the codebase does not currently set a random seed (Section 4), so a single run would not meet this paper's own evaluation protocol. All entries below are placeholders pending a seeded, multi-run evaluation campaign.

6.4. Discussion and Limitations

The one populated result in this section, Table 3, already motivates a methodological caution: because regionalization behavior is network-dependent (Section 6.1), any future closed-loop comparison between regionalization methods must be reported per-scenario, not pooled, since a method that is clearly better on Manhattan (e.g. Leiden's higher modularity) may be indistinguishable from the others on Cologne8 or Ingolstadt21 where all methods nearly collapse to one region.

The remaining, unpopulated tables reflect genuine gaps rather than omissions: Table 4 cannot be populated until the multi-EMV mechanism of Section 3.2 is implemented and at least one scenario's demand file includes an emergency vehicle; Table 5 requires a seeded, multi-run training and evaluation campaign that has not yet been executed under the current PPO-based code. No performance, efficiency, or fairness-improvement claim is made on the basis of either table's placeholder entries. Additional limitations established in this audit — the fixed four-neighbor attention cap (Section 5.4), the gap between the proposed weighted reward design and the reward actually trained on (Section 4), and the absence of random seeding — are consolidated in Section 7.

7. Conclusion

This paper presented FairLight, a hierarchical multi-agent reinforcement learning framework for traffic signal control. Three things are actually implemented and verified against the training code: (i) automatic, graph-based regional partitioning via Louvain, Leiden, DBSCAN, and four hybrid combinations, measured across five SUMO networks in Section 6; (ii) a hierarchical Meta-Policy/ Sub-Policy architecture trained with PPO and GAE, with a fixed, zero-padded four-neighbor graph-attention mechanism at the Sub-Policy level; and (iii) an explicit ordinary-traffic fairness penalty (waiting-time variance, a 90-second starvation threshold, and Jain's Fairness Index) added to the Sub-Policy's local reward. What the experiments in this paper establish is narrower than the full research programme: Table 3 shows that regionalization method choice produces substantially different, network-dependent region structures (from 1 region with near-zero modularity on Cologne8/Ingolstadt21 under Louvain, to 70 regions on Manhattan under Leiden); closed-loop performance comparisons against Fixed-Time and Max-Pressure baselines, and any comparison involving city-wide EMV coordination, remain pending or blocked, respectively, and are not claimed here.

Several limitations currently prevent stronger conclusions. The weighted multi-objective reward of Section 4 ($w_1 R_{\text{eff}} + w_2 R_{\text{fair}} + w_3 R_{\text{emergency}}$) is a design target, not the reward the training code optimizes, which is instead the absolute local reward of Section 3.8 plus the implemented fairness penalty; this gap should be closed before any claim is made about efficiency/fairness/emergency trade-offs under the stated weights. The multi-EMV routing design of Section 3.2 and the incident-robustness mechanism of Section 3.3 are proposed but unimplemented, and no evaluation scenario currently contains an emergency vehicle, so no EMV-related claim — response time, fairness cost of preemption, or otherwise — is currently testable in this codebase. The fixed four-neighbor attention cap (Section 5.4) may discard information at intersections with higher signalized degree; whether this occurs in the five evaluated networks has not been audited. The training and evaluation code does not set a random seed, so the multi-run variability estimates this paper's own evaluation protocol calls for (Section 4) cannot yet be produced. Finally, an earlier, structurally different (pre-PPO) version of this codebase was observed to diverge during training; the current PPO-with-clipping implementation's training stability has not yet been independently re-validated, and this paper makes no stability claim either way for the code described in Sections 3.4–3.10.

Addressing these limitations requires, in order: reconciling the reward design of Section 4 with the implemented training reward (or implementing the gap); adding seed control and executing the Fixed-Time/Max-Pressure/Algorithm evaluation campaign of Section 5 with multiple seeded runs and variability estimates; auditing the five networks' signalized-intersection degree distribution against the four-neighbor attention cap; and, only once an EMV demand profile exists for at least one scenario, implementing and evaluating the multi-EMV mechanism of Section 3.2. Extending validation beyond SUMO toward hardware-in-the-loop or closed-course testing remains a longer-term goal, since SUMO's vehicle models and signal infrastructure are themselves a simplification of an operating real-world traffic system. Section 8 discusses these and related directions in greater detail.

8. Future Works

8.1. Multi-EMV Routing: From Design to Implementation

Section 3.2 sketches, but does not implement, a design in which each active emergency vehicle contributes a time-decaying bias term to regional and intersection-level state representations along its anticipated route, with the Meta-Policy coordinating cross-region handoff and an arbitration rule resolving conflicts between simultaneously active EMVs. Realizing this design requires, in order: (1) adding at least one emergency-vehicle type with origin,

destination, and priority class to a scenario's demand data, since none of the five scenarios evaluated in this paper currently define one; (2) implementing the bias-term injection into the Meta-Policy/Sub-Policy state pipeline; and (3) implementing and evaluating the conflict arbitration rule under at least two simultaneously active EMVs with intersecting corridors. Only once these three steps exist does it become meaningful to compare against EMVLight's dynamic Dijkstra routing [31] or to populate Table 4. The Louvain/Leiden community structure measured in Section 6 may be a useful abstraction for corridor-level planning once this mechanism exists, since inter-community transitions correspond to the sparser boundary links identified by the partitioning step, but this remains an untested hypothesis rather than a demonstrated capability.

8.2. Formal Envy-Free Fairness Optimization

The fairness penalty implemented and evaluated in this paper (Section 4: waiting-time variance, a starvation threshold, and Jain's Fairness Index) does not constitute, and should not be described as, a formal envy-free guarantee: no envy function comparing intersections' or vehicles' allocations against one another is defined or optimized anywhere in the current codebase. A genuine envy-free extension would need to specify, at minimum, which agents' allocations are being compared (e.g. pairs of neighboring intersections, or vehicles sharing an approach), a concrete envy function over a well-defined service quantity, and how a corresponding penalty or constraint would be incorporated into the PPO objective of Section 3.8 without destabilizing training. Only after such a mechanism is implemented would it become appropriate to re-introduce "envy-free" into this framework's name or claims; this paper deliberately avoids that claim given the current, Jain's-index-based fairness mechanism alone.

8.3. Incident and Accident Robustness

Section 3.3 proposes, but does not implement, exposing the controller to injected incidents of varying location, duration, and severity during training, so that the Sub-Policy learns to detect and respond to a sudden capacity loss rather than requiring a hand-engineered override layer. No severity parameter, incident sampling distribution, or incident-aware training loop currently exists in the codebase; the only related functionality implemented is a passive collision counter used for monitoring, not for injecting or training against incidents. Future work should implement an explicit incident-injection mechanism on at least one of the five scenarios of Table 2, then evaluate whether incident-aware training degrades the fairness metrics of Section 4 during incident-free intervals, as originally intended in Section 3.3. Combining incident robustness, multi-EMV routing, and the fairness mechanism already implemented represents the most demanding test of the complete framework and is a plausible target for the next phase of this research, once each component individually reaches the implementation status the others already have.

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